HW2

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Part1

Reading in Data (question 1)

```
beandata <- read_xlsx("C:\\Users\\lbmoy2\\Downloads\\lbmoy2_beans.xlsx")
```

The data has been read into the object `beandata`. This object is a list that has 17 variables and 1 observations.

Modifying beandata by filtering (obs and vars), simplifying var names, and adding new summary variables (question 2)

```
modbean <- beandata %>%
             select(-c(Eccentricity, Solidity)) %>%
               rename("LengthMaj" = MajorAxisLength, "LengthMin" = MinorAxisLength) %>%
                 filter(Class %in% c("CALI", "HOROZ", "SEKER", "SIRA", "DERMASON")) %>%
                   mutate("AvgMaj" = mean(LengthMaj), "AvgMin" = mean(LengthMin) ) %>%
                     mutate("PerimCate" = if_else(Perimeter > 804.048, "Largest", if_else(Perimeter > 7
                       arrange(desc(Extent))

print(as_tibble(modbean))
```

```
## # A tibble: 8,241 x 18
##     Area Perimeter LengthMaj LengthMin AspectRatio ConvexArea EquivDiameter
##    <dbl>     <dbl>     <dbl>     <dbl>       <dbl>      <dbl>         <dbl>
##  1 43185      811.      326.      171.        1.91      43737          234.
##  2 30930      649.      242.      164.        1.47      31278          198.
##  3 49783      911.      375.      171.        2.19      50523          252.
##  4 77815     1076.      433.      231.        1.88      78443          315.
##  5 48282      872.      362.      172.        2.10      48832          248.
##  6 64227     1019.      426.      194.        2.20      64941          286.
##  7 60832      976.      405.      193.        2.10      61584          278.
##  8 56301      961.      403.      180.        2.24      57012          268.
##  9 55019      931.      391.      181.        2.16      55802          265.
## 10 42678      823.      341.      161.        2.12      43227          233.
## # i 8,231 more rows
## # i 11 more variables: Extent <dbl>, Roundness <dbl>, Compactness <dbl>,
## #   ShapeFactor1 <dbl>, ShapeFactor2 <dbl>, ShapeFactor3 <dbl>,
## #   ShapeFactor4 <dbl>, Class <chr>, AvgMaj <dbl>, AvgMin <dbl>,
## #   PerimCate <chr>
```

Making 2-way Contingency table (question 3)

```
modbean %>%
  filter(LengthMaj < 404.4919)
```

```
## # A tibble: 7,416 x 18
```

```
##     Area Perimeter LengthMaj LengthMin AspectRatio ConvexArea EquivDiameter
##    <dbl>     <dbl>     <dbl>     <dbl>       <dbl>      <dbl>         <dbl>
##  1 43185      811.      326.      171.        1.91      43737          234.
##  2 30930      649.      242.      164.        1.47      31278          198.
##  3 49783      911.      375.      171.        2.19      50523          252.
##  4 48282      872.      362.      172.        2.10      48832          248.
##  5 56301      961.      403.      180.        2.24      57012          268.
##  6 55019      931.      391.      181.        2.16      55802          265.
##  7 42678      823.      341.      161.        2.12      43227          233.
##  8 32915      682.      263.      160.        1.64      33250          205.
##  9 55333      933.      391.      181.        2.16      55814          265.
## 10 52385      856.      325.      206.        1.58      52904          258.
## # i 7,406 more rows
## # i 11 more variables: Extent <dbl>, Roundness <dbl>, Compactness <dbl>,
## #   ShapeFactor1 <dbl>, ShapeFactor2 <dbl>, ShapeFactor3 <dbl>,
## #   ShapeFactor4 <dbl>, Class <chr>, AvgMaj <dbl>, AvgMin <dbl>,
## #   PerimCate <chr>
```

```
table(modbean$Class, modbean$PerimCate)
```

```
##
##            Largest Lowest Medium
##   CALI        1152      0      1
##   DERMASON       1   1916    552
##   HOROZ       1237      3     79
##   SEKER        103    520    807
##   SIRA         803     33   1034
```

The top left number is the number of cali beans that have the largest perimeter.

Numeric Summary (question 4)

```
modbean %>%
  group_by(Class) %>%
    summarize(AvgAr = mean(Area), AvgPer = mean(Perimeter),  MinAr = min(Area), MinPer = min(Perimeter)
            SkewAr = skewness(Area), SkewPer = skewness(Perimeter) , VarAr = var(Area), AvgPer = var(Pe
```

```
## # A tibble: 5 x 8
##   Class     AvgAr AvgPer MinAr MinPer  SkewAr SkewPer     VarAr
##   <chr>     <dbl>  <dbl> <dbl>  <dbl>   <dbl>   <dbl>     <dbl>
## 1 CALI     75520.  4730. 45666   790.  0.445   0.150  91721871.
## 2 DERMASON 32088.  2529. 20464   525.  0.103   0.0202 21712828.
## 3 HOROZ    53726.  5053. 33006   689. -0.0607 -0.210  55063780.
## 4 SEKER    39912.  2281. 28395   610.  0.703   0.762  22916372.
## 5 SIRA     44741.  2025. 31519   668.  0.148   0.146  21194880.
```

The cali bean has an avg area of 75519.73, avg perimeter of 4730.073, a min area of 45666 and min perimeter of 789.770. Its skewness of the area is 0.44478747 and the skewness of the perimeter is 0.1495999 (both are more positively skewed). The variance is 91721871.

Part2

URL : https://archive.ics.uci.edu/dataset/42/glass+identification

Why it interests me: I just picked this out because it said glass and I think reflections are cool. The contents don't interest me too much, nor do I know why this data is important.

Reading in Data

```
glassdata <- read_csv("C:\\Users\\lbmoy2\\Downloads\\glass+identification\\glass.data", col_names = FALS
  rename("Id" = X1,
         "Refrac_index" = X2,
         "Sodium" = X3,
         "Magnesium" = X4,
         "Aluminum" = X5,
         "Silicon" = X6,
         "Potassium" = X7,
         "Calcium" = X8,
         "Barium" = X9,
         "Iron" = X10,
         "Type_of_glass" = X11)
```

Two questions (that can be answered) 1) What is the max refraction index?

2) How do the averages of elements compare to one another?

Summary Stats and Tables

```
glassdata %>%
  group_by(Type_of_glass) %>%
    summarize( max = max(Refrac_index))
```

```
## # A tibble: 6 x 2
##   Type_of_glass   max
##           <dbl> <dbl>
## 1             1  1.53
## 2             2  1.53
## 3             3  1.52
## 4             5  1.52
## 5             6  1.52
## 6             7  1.52
```

```
glassdata %>%
  summarize(smean = mean(Sodium), mmean = mean(Magnesium),amean = mean(Aluminum),simean = mean(Silicon)
```

```
## # A tibble: 1 x 8
##   smean mmean amean simean pmean cmean bmean  imean
##   <dbl> <dbl> <dbl>  <dbl> <dbl> <dbl> <dbl>  <dbl>
## 1  13.4  2.68  1.44   72.7 0.497  8.96 0.175 0.0570
```

1)Max Refraction Index is from type 2 glass with 1.534 2) Silicon is obviously has the highest concentration in glass and the next is sodium.